Chapter 16 | Written solutions

From Pretrained Weights to Task Behavior

All twelve exercises are solved here, from counting a classifier to checking an adapter checkpoint and planning a fair comparison. The key distinctions are between fixed weights and gradient paths, input context and loss targets, and relative preferences and absolute response probabilities. Every numerical claim below is checked using small CPU arrays or tensors.

The written PDF retains every derivation, result and interpretation. The notebook PDF also includes the code and saved outputs. Synthetic examples verify the mechanisms; they do not measure real-world adaptation quality.

1. Count a task head

Each of three class scores uses six weights, one per input feature, and one bias. Thus the head has $3 \cdot 6 + 3 = 21$ trainable scalars. Head-only training updates 21 scalars; full fine-tuning updates $120 + 21 = 141$. Both systems still need the encoder to make predictions. Freezing changes the update budget, not the existence or storage of the base weights.

2. Mark the response targets

Write each target underneath the input whose logits predict it. Use zero-based indices so the table also matches tensor indexing.

Logit row	Input token	Next target	Loss weight
0	BOS	Question	0
1	Question	Assistant	0
2	Assistant	New	1
3	New	Address	1
4	Address	EOS	1
5	EOS	PAD	0

The response starts with `New`, so the first supervised prediction comes from the prompt's `Assistant` row. `EOS` is a supervised target because the model must learn when to stop. Its own logit row predicts padding and is ignored. The final supplied padding token has no next target in this sequence. There are six shifted pairs and three supervised targets.

The loss mask selects targets. Causal attention still lets response predictions use the prompt, including positions whose next targets are ignored. The response loss can backpropagate through those context paths.

3. Follow a rank-one update

The base is the identity, so $W_0\mathbf{x} = (1, 2)^\top$. The down-projection forms $A\mathbf{x} = 2 \cdot 1 + 1 \cdot 2 = 4$. The up-projection gives $(4, -4)^\top$, and the scale $\alpha/r = 0.5$ makes the correction $(2, -2)^\top$. Adding it to the base output gives $(3, 0)^\top$. To merge, first form the outer product and then add the scaled update:

$$BA = \begin{bmatrix} 2 & 1 \\ -2 & -1 \end{bmatrix}, \qquad W_{\text{merged}} = I_2 + 0.5BA = \begin{bmatrix} 2 & 0.5 \\ -1 & 0.5 \end{bmatrix}.$$

Multiplying this matrix by $(1, 2)^\top$ again gives $(3, 0)^\top$. The correction has rank one, while the merged matrix has rank two. LoRA restricts the directions available to the update; the original map still contributes its own directions.

4. Compute a relative preference

The policy gap is $-2 - (-3.5) = 1.5$ and the reference gap is $-3 - (-3.25) = 0.25$. With $\beta = 1$, the margin is $m = 1.25$ and

$$L = -\ln \sigma(1.25) = \ln(1 + e^{-1.25}) \approx 0.251929.$$

After subtracting 0.5 from both policy scores, the scores are $(-2.5, -4)$. The gap is still 1.5, so the margin and loss stay the same. Both response probabilities were multiplied by $e^{-0.5} \approx 0.606531$, but their ratio did not change. A complete distribution can assign the freed probability to other responses. This is a statement about this pair's relative loss, not a guarantee about how a shared neural model will update every response.

5. Check target alignment in code

Suppose the model has read `BOS Question Assistant` and is about to predict `New`. This prediction belongs to the logit row at `Assistant`. For an input row at position t, the next-token target comes from position $t + 1$. The mask first marks the **tokens to predict** in their original positions. The helper then shifts both the token IDs and the mask so that each target lines up with the preceding logit row.

Number the seven positions from zero. Use a long tensor with batch size one and mark positions 3, 4, and 5 as the response. The end marker `EOS` is part of the response because the model must learn when to stop. Padding at position 6 is excluded.

Logit row	Input token	Next token and ID	Surviving label
0	`BOS`	`Question` (3)	`IGNORE`
1	`Question`	`Assistant` (4)	`IGNORE`
2	`Assistant`	`New` (5)	5
3	`New`	`Address` (6)	6
4	`Address`	`EOS` (2)	2
5	`EOS`	`PAD` (0)	`IGNORE`

The helper uses `IGNORE = -100` as a label sentinel. It is outside the vocabulary and means this row contributes no supervised target. There is no target after the last supplied token, so the result has shape $[1, 6]$. The scoring helpers use the first six logit rows when given the original seven-row logits.

Notice that `Assistant` is a prompt token but its logit row is scored because the **next** token is `New`. Conversely, `EOS` is a response token but its own logit row is ignored because the next token is padding. Selecting targets and providing input context are separate operations.

6. Perturb a logit row

Keep the same labels. Give the model seven logit rows with seven vocabulary entries in each row. All entries start at zero, so softmax assigns probability $1/7$ to every token. Only the three targets at rows 2, 3, and 4 count. The mean loss is $\ln 7$, and the completion log-probability is the **sum** $-3 \ln 7$ for this one sequence.

First raise the `Question` score in row 0 (`BOS`) from zero to 2. This changes the row's probability of `Question`, but `Question` is an ignored target. The probability distribution at every other row stays the same.

For the second intervention, start again from all zeros and raise the `New` score in row 2 (`Assistant`) to 2. Row 2 now assigns `New` probability

$$p(\texttt{New}) = \frac{\exp(2)}{\exp(2) + 6}.$$

There are still six other vocabulary entries with score zero in that row. The negative log-probability of `New` becomes

$$d = -\ln p(\texttt{New}) = \ln(\exp(2) + 6) - 2 = \ln(1 + 6\exp(-2)).$$

Rows 3 and 4 still contribute $\ln 7$ each. Thus the new mean loss is $(d + 2\ln 7)/3$ and the new completion log-probability is $-d - 2\ln 7$. The denominator of the mean stays three because neither intervention changes the labels.

Verify the two formulas independently. A mean loss and a summed log-probability have different units of aggregation. Here there is one sequence with exactly three scored positions, so multiplying its mean loss by -3 recovers the completion log-probability. This scaling would not turn a mean over a mixed batch into one sum per response.

Raising the correct score in a selected row makes the mean loss smaller and the summed log-probability less negative. The ignored-row experiment changes already-computed logits. During a model forward pass, an ignored prompt token can still influence later response logits through causal attention.

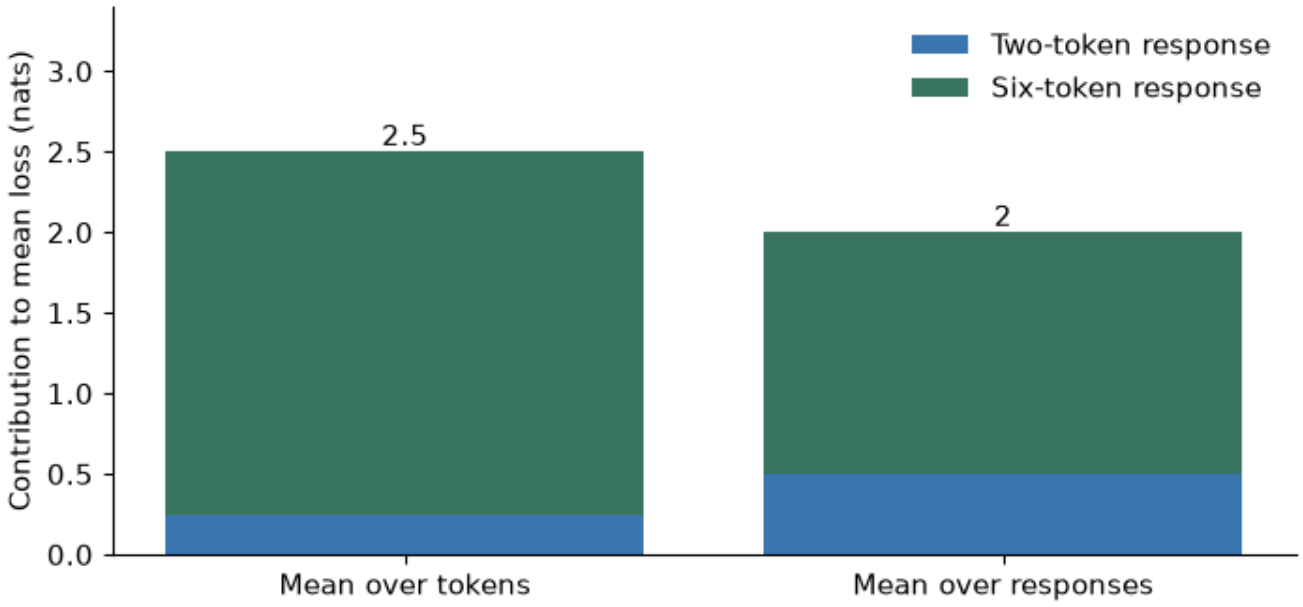
7. Compare two batch averages

The short response contributes total loss 2 from two tokens. The long one contributes total loss 18 from six tokens. A mean over all supervised tokens is $(2 + 18)/(2 + 6) = 2.5$. A mean of response means is $(2/2 + 18/6)/2 = 2$.

For vocabulary IDs 0 and 1, put logits $(\ln(e^\ell - 1), 0)$ at a selected row with target ID 1. Its probability is $1/[(e^\ell - 1) + 1] = e^{-\ell}$, so its negative log-probability is exactly ℓ. Use $\ell = 1$ for the first sequence and $\ell = 3$ for the second. The first token in each row supplies context; it is not a scored target.

We keep the padding ID distinct from selected targets. Padding still has a vocabulary logit and participates in the softmax denominator at a selected row; it simply never serves as a selected target. The labels align with the preceding logits after shifting.

Adding right padding supplies no additional supervised target. Recompute the labels and both reductions to verify that each answer is unchanged.



Under the token mean, the long response has six of eight token slots; the short one has two. Under the response mean, each response's own mean gets weight one half. The tall green segment also reflects its larger loss per token. A reported loss is meaningful only with its target mask and reduction.

8. Inspect a fresh adapter

Imagine adding a second path to an existing four-input, three-output linear layer. The original path keeps its weight and bias. The new path first compresses the input from four features to two, then expands it to three. This is low-rank adaptation, or LoRA. A fresh adapter is built so this second path starts by adding zero.
Write the base weight as W_0, its bias as $\mathbf{b}_0$, and the two LoRA factors as A and B. The rank setting is $r = 2$ and the scale is $\alpha/r = 4/2 = 2$. Using column vectors, the computation is

$$\mathbf{y} = W_0\mathbf{x} + \mathbf{b}_0 + 2B(A\mathbf{x}).$$

The factor A has shape $[2, 4]$, so $A\mathbf{x}$ has two entries. The factor B has shape $[3, 2]$, so the update has three entries and can be added to the base output. The implementation initializes A randomly and B to zero. Consequently the update is the zero vector even if $A\mathbf{x}$ contains nonzero entries. The base bias still contributes to the output.

Create the input with `base.weight.new_tensor` so its dtype and device match the layer. This run uses CPU float64. The seed fixes the illustrative parameter values and the numeric output. The equality at initialization follows from $B = 0$ for any base initialization.
Inspect all four parameter tensors. A `requires_grad` flag of `True` makes a parameter eligible to receive gradients. This wrapper sets the base flags to `False` and leaves both factors trainable.

Parameter	Shape	Number of scalars	Trainable
Base weight W_0	$[3, 4]$	12	No
Base bias $\mathbf{b}_0$	$[3]$	3	No
Factor A	$[2, 4]$	8	Yes
Factor B	$[3, 2]$	6	Yes

There are $2 \cdot 4 + 3 \cdot 2 = 14$ trainable scalars and $15 + 14 = 29$ scalars in the wrapper altogether. The zero-initialized factor still counts as trainable. Its stored values are zero now, while its gradient flag permits an optimizer to change them.

The scalar extension shows how learning starts. With $y = 2 + 4ba$ and $L = y^2/2$, the gradients are $\partial L/\partial b = 4ya$ and $\partial L/\partial a = 4yb$. Initially $a = 1$, $b = 0$, so $y = 2$, $\partial L/\partial b = 8$, and $\partial L/\partial a = 0$. One SGD step of size 0.1 gives $b = -0.8$ and $a = 1$. Now $y = -1.2$, the loss is 0.72, and the gradients are -4.8 for b and 3.84 for a.

If both factors start at zero, the output still equals the base output, but both factor gradients are zero. Ordinary SGD cannot start changing this branch. A zero initial *update* is useful; zeroing every factor is not.

9. Compare trainable weights

Train two versions of the same small classifier on the same eight examples. The frozen version lets only the task head change. The full version also lets the encoder learn. For each version, run one experiment with one optimizer step and a separate experiment with two. In particular, the two-step run starts over and does not resume from a trained model.

Save a snapshot of one untouched `TinyClassifier` and load it into four fresh instances. All four therefore start with the same encoder and the same zero head. Use only the first two outputs of `classification_data`, the training inputs and class labels. The helper trains with full-batch SGD, meaning one update uses all eight examples, and we give it the same learning rate of 0.2 in every run. Holding these choices fixed means that the encoder's trainable status and the number of updates are the only differences in the comparison.

For each weight and bias tensor, subtract its original value and take the largest absolute entry in that difference. A zero means the whole tensor is unchanged. Float64 comparisons use absolute tolerance 10^{-12} and zero relative tolerance. The results distinguish changes from numerical residue in the bias entries.

Why do both encoders stay put on step 1? Let $\mathbf{a}_i = E\mathbf{x}_i + \mathbf{c}$ be the encoder's pre-activation vector and $\mathbf{h}_i = \tanh(\mathbf{a}_i)$ its feature vector for example i. The head weight H and bias $\mathbf{b}$ give logits $\mathbf{z}_i = H\mathbf{h}_i + \mathbf{b}$. Let $\mathbf{g}_i$ be the cross-entropy gradient for example i with respect to its logits before batch averaging. The gradient components are the predicted probabilities minus the one-hot target.

At initialization, every logit is zero. The two predicted probabilities are both $1/2$, so the target class has gradient $-1/2$ and the other class has gradient $1/2$. These nonzero gradients reach the head weight, even though its entries are currently zero. For a batch of m inputs the weight gradient is an average of outer products,

$$\nabla_H L = \frac{1}{m}\sum_{i=1}^{m} \mathbf{g}_i \mathbf{h}_i^\top.$$

The encoder receives a different gradient. Backpropagation multiplies the head's logit gradient by the transpose of its weight, then applies the elementwise derivative of $\tanh$. With $\odot$ denoting elementwise multiplication, the gradient of the mean loss at the encoder pre-activation is

$$\nabla_{\mathbf{a}_i} L = \frac{1}{m}(H^\top \mathbf{g}_i) \odot (1 - \mathbf{h}_i^2).$$

At the first backward pass $H = 0$, so the encoder gradient vanishes even in the full run, where those weights have `requires_grad=True`. SGD then updates H. On the second backward pass the nonzero head weight passes a gradient to the encoder. The full encoder weight changes at that step, and the frozen encoder remains unchanged. The frozen weight has no gradient stored, while the trainable encoder's first gradient is an actual tensor of zeros.

Both head weights agree after one step and after two. At the start of step 2, the head and encoder still have the same values in both runs. The head's gradient is therefore the same in that backward pass, before the optimizer changes the full encoder. If training continues, later head updates can differ because the encoders now supply different features.

The first head-bias gradient sums to zero because the labels are balanced and the initial predictions are uniform. The tiny bias differences in the second-step runs are at the scale of floating-point rounding in this symmetric example. Comparing the weight tensors separately avoids mistaking those tiny entries for substantial changes.

An independent calculation from these matrix products reproduces the step-1 head update and step-2 encoder update without calling backward. After one step, both encoder weights are unchanged and the largest head weight change is about 0.08237. After two steps, the largest head change is about 0.1568 in both runs. The frozen encoder remains unchanged; the full encoder's largest weight change is 0.008305283. The bias changes are below the stated tolerance in this symmetric example.

Now isolate a different issue: a frozen operation between a trainable parameter and the loss. Set $h = ux$, $y = 3h$, and $L = y^2/2$, with $u = 1$, $x = 2$. The multiplier has no trainable gradient, but the gradient reaching u is $y \cdot 3x = 36$. A frozen weight remains part of the chain rule.

Placing the multiplication inside `no_grad` gives an output with no tracked connection to u. Here the resulting loss cannot call backward at all; in a larger model with a later trainable head, that head could still train while the path to u would remain broken. `eval()` is different again: it selects evaluation behavior of modules, without globally disabling autograd.

10. Keep the reference fixed (optional)

Direct Preference Optimization (DPO) compares the policy's preference for a chosen response over a rejected response with the preference already expressed by a reference model. The policy gap is $-2-(-3.5)=1.5$. The reference gap is $-3-(-3.25)=0.25$. With $\beta=1$ the margin is

$$m=\beta\big[(p_c-p_r)-(r_c-r_r)\big]=1.25.$$

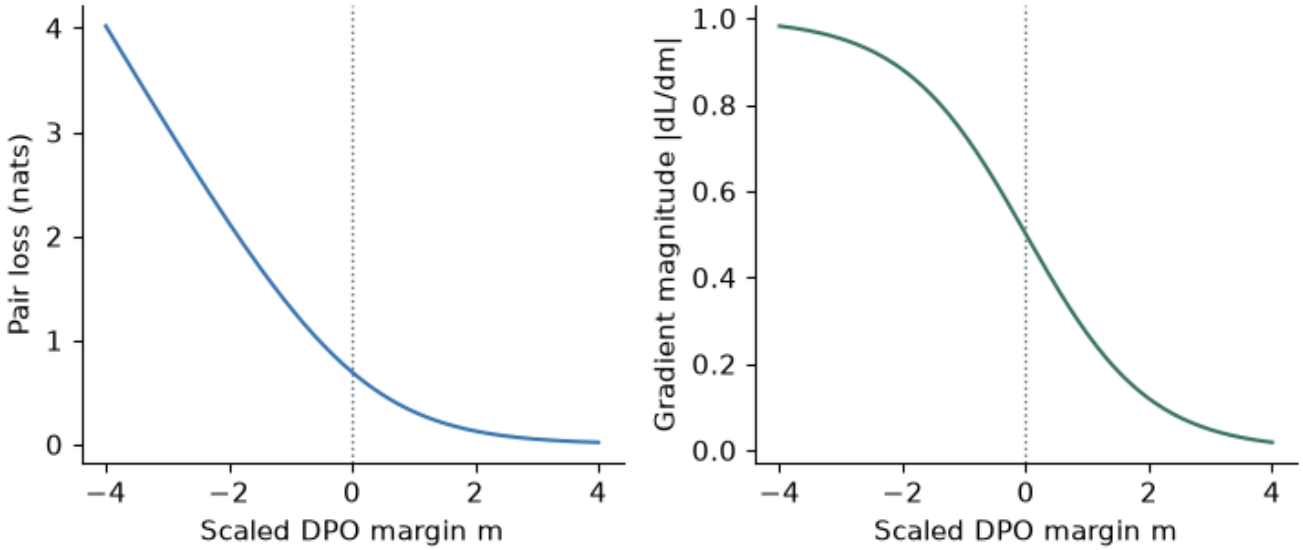
Here p_c, p_r denote policy log-probabilities and r_c, r_r denote reference log-probabilities. For one pair the loss is $L=-\ln\sigma(m)=\ln(1+\exp(-m))$, where $\sigma(z)=1/(1+\exp(-z))$. Its derivative with respect to the margin is

$$\frac{dL}{dm}=-\frac{1}{1+\exp(m)}=-\sigma(-m).$$

The policy scores enter the margin with derivatives $+\beta$ for the chosen response and $-\beta$ for the rejected response. Their loss gradients are therefore $-\beta\sigma(-m)$ and $+\beta\sigma(-m)$. If the scores could be updated directly, gradient descent would raise the chosen score and lower the rejected one. In a model these gradients flow through the response log-probabilities to shared trainable weights.

In the code, the reference scores are detached before their difference is computed. They supply the value 0.25 for the forward pass, while autograd has no path from the loss back to their original tensors. Give all four tensors `requires_grad=True` so the test distinguishes detachment from simply starting with gradients disabled.

Both reference gradient fields stay `None`. In this fresh backward pass, `None` means no gradient was accumulated for that tensor. A gradient tensor containing zero would instead represent a computed value. Detachment affects this loss's graph without changing the original reference tensors' `requires_grad` flags.



The loss decreases as the relative margin increases. A poorly ranked pair supplies a larger gradient with respect to that margin. The plot is the scalar loss function, not measured model performance. When gradients flow into shared weights, different pairs and tokens can interact.

11. Rebuild an adapted model (optional)

Save both factors A and B for every adapted query and value projection, plus the new classifier's weight and bias. Identify each factor by its full module path and record its shape, orientation, rank, alpha, and scaling rule. Include any other trained biases or modules. Save the immutable base revision and configuration, tokenizer and its revision, special-token IDs, prompt format, label order, pooling rule, storage dtype, and evaluation settings. An adapter trained for one base is not a self-contained replacement for it.

For a training restart, additionally save optimizer and scheduler state, training progress, random-generator states and the data position. These restart states are distinct from the inference checkpoint.

For a two-input affine layer, test zero, $(1, 0)^\top$, and $(0, 1)^\top$. The zero input gives the bias. Subtracting that result from each unit-vector output gives the corresponding matrix column. Together these probes check the full affine map. The example below restores factors onto a known base and compares the separated and merged paths, including the base bias.

The numerical fixture uses $W_0 = \left[\begin{smallmatrix} 1 & 2 \\ 0 & -1 \end{smallmatrix}\right]$, $A = (1, -1)$, $B = (0.5, 1)^\top$, scale 2, and base bias $(0.25, -0.5)^\top$. The merged matrix is $\left[\begin{smallmatrix} 2 & 1 \\ 2 & -3 \end{smallmatrix}\right]$. The zero, first-coordinate and second-coordinate probes respectively produce $(0.25, -0.5)$, $(2.25, 1.5)$ and $(1.25, -3.5)$. Both the restored-factor calculation and the merged layer match all three outputs to absolute tolerance 10^{-12}.

One probe can miss a nonzero error. Add $D = \left[\begin{smallmatrix} 1 & -1 \\ 0 & 0 \end{smallmatrix}\right]$ to a weight matrix. The error vanishes on input $(1, 1)^\top$, since $D(1, 1)^\top = 0$. Both coordinate probes detect it. This also explains why an input that barely activates a swapped projection might fail to expose a loading mistake.

A query and a value projection can have identical shapes, so check their names and their numerical effects separately. Use a probe that activates each saved adapter. Here both adapters have rank one and scale two, but at the first coordinate input their corrections are $(1, 2)^\top$ for query and $(2, -1)^\top$ for value. Swapping the factors produces a visible mismatch. The adapted-minus-base difference cancels the base term: correct adapter corrections alone therefore cannot verify the base checkpoint.

For the complete attention model, check base identifiers, exact projection names and shapes before loading. Compare saved pre-reload logits with post-reload logits on a varied fixed batch, using the same tokenizer, formatting, padding, label order, dtype and evaluation mode. Include inputs of different lengths and examples that exercise each class. Check each adapted projection's intermediate output when diagnosing a mismatch.

Use tolerances appropriate to precision and quantization. Finite output checks are evidence of agreement on those inputs; they do not prove that two nonlinear models agree on every input. Configuration and identity checks remain necessary even if a small batch passes.

12. Plan an adaptation comparison (optional)

Give all three systems the same encoder architecture, tokenizer, pooling rule, and two-class head returning `LOOKUP` or `UPDATE`. Use the same pretrained base revision and same initial head within each paired run. The conditions are: head only; full base and head; and LoRA on declared projections plus the head. One explicit starting configuration is rank four, alpha eight, and adapters on every query and value projection. Their zero initial update preserves the initial encoder function. Record dropout behavior and keep it consistent where the comparison requires the same stochastic model. Also report the unchanged pretrained classifier when one exists, or a clearly specified zero-training prompting baseline with its different output interface reported separately. A majority-class baseline provides an additional task-level reference.

Group messages by their generating template and customer before splitting. If those grouping relationships overlap, keep the connected families together. Split groups into training, validation and final test sets; keep paraphrases and near-duplicates in the same split. If deployment is meant to generalize to later time periods or new customers, reflect that in the split. Check for duplicates before tuning any configuration.

Keep data, labels, preprocessing and evaluation fixed. Give each method a declared tuning budget and use validation data to select learning rate, rank and stopping time. Repeating the same learning rate for every method is a controlled diagnostic, but may not compare each method fairly at its best setting. Report the selected configuration, examples seen and compute used. Repeat over several predetermined seeds, pairing initial heads and data order across methods when possible. Report per-run results and spread, and use the test set only for the final comparison. Check that each split contains both classes. Include contrastive phrasings such as asking to confirm the new address versus asking to change to it, so that the words "new address" do not uniquely identify a class. If estimating sampling uncertainty, resample whole test families to preserve their within-group dependence.

Measure test accuracy, the confusion matrix, and per-class precision and recall. Inspect errors on unfamiliar phrasings. If preserving earlier skills matters, evaluate those too. For a generated-label baseline, separately measure valid output format and correct labels under the same parsing rule. Do not count malformed outputs as correct without disclosing that rule.

Report trainable and total parameter counts, per-task checkpoint bytes, and the base storage required to use each checkpoint. Measure peak training memory and elapsed time on specified hardware with the same precision, batch size and sequence lengths. Parameter counts alone do not determine activation memory, optimizer state, runtime or measured accuracy. State whether the deployed adapter is merged: merging removes the separate factor tensors from the deployed linear layers. Distinguish raw tensor payload from actual file size, which also depends on metadata and compression.

For a concrete storage calculation, suppose the 120-parameter encoder contains one selected 6×6 matrix and uses the three-class head from Exercise 1. A rank-two adapter adds $2(6 + 6) = 24$ scalars. The following counts exclude file-format metadata and use four bytes per scalar. This is a counting illustration, separate from the two-class address experiment.

Method	Trainable scalars	Total resident scalars	Per-task weight bytes
Head only	21	141	84
Full fine-tuning	141	141	564
LoRA plus head	45	165	180

Head and adapter checkpoints both need the original 120-scalar base, which takes another 480 payload bytes if stored in float32. The full checkpoint already contains its updated base. Adam state or mixed-precision copies would add training storage; those are not included in this table. No accuracy ranking follows from these counts.